

A cross-sectional study of oxidative stress pathway genotypes and their interactions with environmental pollutant levels identifies associations with gene expression and lung function



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Summary

Background Asthma is a heterogeneous disease influenced by genetic and environmental factors. Fine particulate matter (PM_{2.5}) exacerbates asthma, likely through oxidative stress pathways, but whether genetic variation modifies this effect remains unclear.

Methods We analysed data on 948 adults with asthma from the Severe Asthma Research Program (SARP), linking ZIP-code-level PM_{2.5} exposure with whole-genome sequencing data. We tested 4337 single nucleotide polymorphisms (SNPs) in 120 oxidative stress pathway genes for gene–environment (GxE) interactions with PM_{2.5} on lung function (forced expiratory volume in 1 s [FEV₁] % predicted) using weighted linear regression. Gene expression data from bronchial epithelial cells (n = 170) were used to assess cis-expression quantitative trait loci (eQTLs).

Findings Higher PM_{2.5} exposure was associated with lower FEV₁% predicted (β per $\mu\text{g}/\text{m}^3 = -0.7$, $p = 0.01$). We identified 20 SNPs across seven genes (*OXSRI*, *PXDN*, *TPO*, *LRRK2*, *APP*, *MSRA*, *MSRB2*) with significant GxE interactions after multiple-testing correction. Five SNPs were also eQTLs, linking PM_{2.5}-modified gene expression to lung function. Minor alleles in *OXSRI* and *PXDN* were associated with reduced gene expression and worsened FEV₁% under high PM_{2.5} exposure. Conversely, *TPO* variants were associated with higher baseline expression and lower lung function, but under increasing PM_{2.5} exposure, minor allele carriers showed suppressed *TPO* expression and improved FEV₁%.

Interpretation This study identified 20 SNPs in oxidative stress pathway genes that modify the effect of PM_{2.5} on lung function in asthma. These findings highlight the importance of integrating environmental context in genetic studies and suggest potential therapeutic targets for pollution-sensitive asthma phenotypes.

Funding Supported by NIH grants.

eBioMedicine
2026;129: 106334

Published Online 23 June
2026

<https://doi.org/10.1016/j.ebiom.2026.106334>

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Keywords: Pollution; Asthma; Genexenvironment; PM_{2.5}; Spirometry

Research in context

Evidence before this study

Asthma is a complex and heterogeneous disease shaped by both genetic and environmental influences. Air pollution, particularly fine particulate matter (PM_{2.5}), is a well-recognised risk factor for airway obstruction and asthma exacerbations, with oxidative stress pathways thought to mediate much of its harm. Prior gene–environment (GxE) studies have largely focused on a small number of candidate genes, such as the glutathione S-transferase family, and on childhood asthma susceptibility, rather than lung function in adults. Moreover, these studies were often underpowered, with limited sample sizes and genetic resolution, leaving a gap in our understanding of how individual genetic variation in oxidative stress pathways modifies the respiratory response to air pollution.

Added value of this study

This study represents one of the most comprehensive GxE analyses of lung function in asthma to date, combining whole-genome sequencing, ZIP code-level PM_{2.5} exposure, and bronchial epithelial gene expression profiling in a well-characterised cohort of 948 adults with asthma. By systematically evaluating 4337 SNPs in 120 oxidative stress pathway genes, we identified 20 SNPs across seven genes (*OXSRL1*, *PXDN*, *TPO*, *LRRK2*, *APP*, *MSRA*, *MSRB2*) that significantly interacted with PM_{2.5} exposure to influence lung

function. Furthermore, five SNPs also functioned as cis-expression quantitative trait loci, supporting biologically plausible mechanisms through altered gene expression. These findings not only expand the catalogue of pollution-sensitive genetic variants in asthma, but also link them to functional gene regulation in airway epithelium.

Implications of all the available evidence

The findings of this study advance our understanding of the molecular mechanisms underlying interindividual susceptibility to air pollution in asthma. The identified gene–environment interactions highlight oxidative stress pathways as potential mediators of pollution-related lung function impairment and identify specific genes that may serve as targets for therapeutic intervention. These results underscore the necessity of incorporating environmental exposures into genetic and genomic investigations of asthma and support the potential utility of genetic markers to inform risk stratification. Future research should aim to replicate these findings in independent populations, elucidate the causal pathways linking genotype, gene expression, and lung function, and assess whether targeted modulation of oxidative stress pathways can mitigate the adverse effects of air pollution on lung function in susceptible individuals.

Introduction

Asthma is a chronic inflammatory disorder of the airways characterised by bronchial hyper-responsiveness and reversible airflow limitation.¹ Forced expiratory volume in 1 s (FEV₁) is a key physiologic parameter that assesses airflow limitation, and is an important tool for both asthma diagnosis and assessment of control and severity.¹ Asthma is also now recognised as a highly heterogeneous disease for which both genetic and environmental risk factors are involved.² Genetic factors related to airway structure, T helper type 2 (Th2) immunity, the HLA locus, and oxidative stress pathways have been reported to predispose individuals to greater airflow limitation (airway obstruction).³ Patients with asthma appear to be particularly sensitive to air pollution, as exposure increases airway obstruction both acutely and chronically while worsening asthma control and severity.⁴

Particulate matter, less than or equal to 2.5 µm diameter (PM_{2.5}) exposure has also been reported to be associated with lowered lung function in patients with

asthma.⁴ Airborne pollutants can originate from anthropogenic sources, such as vehicular exhaust, industrial sources, and coal combustion in urban centres, as well as from organic sources like biomass burning and agricultural activities in rural areas.⁵ Gaseous pollutants including ozone (O₃), nitrogen dioxide (NO₂), carbon monoxide (CO), and sulphur dioxide (SO₂) can combine with PM to form mixes of solids, liquids, and vapours that can be inhaled.⁶ As these pollutants penetrate deep into the respiratory system, they generate reactive oxygen species (ROS) through chemical components or metal-mediated Fenton reactions, leading to persistent oxidative stress and inflammation. These changes have been reported to worsen airway hyperresponsiveness, remodelling, and airway obstruction.⁴

Enhancement of oxidative stress pathways likely plays a central role in the biologic and clinical effects of PM exposure. Previous studies have evaluated single nucleotide polymorphisms (SNPs) in selective candidate genes for their potential to drive GxE

interactions.^{7,8} SNPs in the glutathione S-transferase family have been extensively investigated for their interactive effects with traffic-related air pollutants in childhood asthma onset.^{7,9} Interactions of PM₁₀ exposure with SNPs in oxidative stress pathway genes (*NFE2L2* (Nrf2), superoxide dismutase) have also been reported with hospital admissions.¹⁰ Although GxE interactions have been studied in asthma development and exacerbations, evidence linking GxE to lung function in asthma patients is limited, in contrast to well-established findings in COPD and occupational exposure.^{11,12} These previous studies have also been constrained to a small number of candidate genes.¹³

The aim of this study was to comprehensively investigate whether known polymorphisms in a wide range of oxidative stress response pathway genes modulate the effects of PM_{2.5} on lung function (specifically FEV₁% predicted) in patients from the well-characterised Severe Asthma Research Program (SARP). We then incorporated airway epithelial brushing gene expression profiles to explore whether these polymorphisms may have influenced lung function by serving as expression quantitative trait loci (eQTLs) for gene expression. The objective of this study was to investigate how genetic variation in oxidative stress-related pathways modifies the association between long-term PM_{2.5} exposure and lung function in adults with asthma.

Methods

Study subjects

SARP is a multi-centre study to investigate the pathobiology of asthma phenotypes using standardised protocols. The initial cohort, encompassing SARP 1 and 2, was cross-sectional and included 1685 individuals. The subsequent cohort, SARP 3, was designed as a longitudinal study and included 763 individuals. Sex was self-reported at enrolment and included as a biological variable in regression models. No sex-based exclusions were applied. Only participants with both 5-digit zip codes (US only) ($n = 1457$, excluding $n = 740$ for missing zip code data) and whole genome sequencing (WGS) ($n = 1688$, excluding $n = 760$ for missing genotype data) were included in the analysis (total $n = 948$) (see Flowchart in Fig. 1). As this was a convenience sample derived from available SARP participants with complete WGS and exposure data, no formal sample size calculations were performed a priori. The cohort was further categorised into traditional industrialised and non/less-industrialised sites. Cleveland, St. Louis, and Pittsburgh were considered industrialised, while Boston, Denver, Galveston, Madison, San Francisco, and Winston-Salem were considered less industrialised. FEV₁% predicted values were calculated using the GLI Global 2022 reference equations, which provide race-neutral predicted values.¹⁴

Ethics

All research activities were conducted with the approval of the respective institutional review boards at the participating research sites, and informed consent was obtained from all study participants. This study was conducted as part of the Severe Asthma Research Program IV (SARP IV): Immunometabolic phenotypes in adult severe asthma and disease progression. The protocol was reviewed and approved by the Institutional Review Board (IRB) at Wake Forest University Health Sciences under protocol number IRB00066576. All study participants provided written informed consent prior to participation in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments.

Genetic and genomic studies

WGS was performed in SARP (Freeze 6; dbGaP accession: phs001446.v3.p2) through the NHLBI Trans-Omics for Precision Medicine (TOPMed) program, primarily using Illumina HiSeq X Ten platforms with PCR-free library preparation and $\geq 30 \times$ coverage.¹⁵ Bronchial epithelial cell (BEC) gene expression was profiled from RNA extracted via bronchoscopic airway brushings in asthma patients. SARP 1–2 Microarray gene expression (GSE63142) was processed using Agilent workflows.^{16,17} Background correction and probe filtering were performed, followed by quality control using array-wide intensity distributions, MA plots, and detection of outlier samples. Expression values were log₂-transformed and normalised using cyclic-LOESS with Bioconductor, as documented in the original SARP dataset.

SARP 3 RNA-seq data underwent standard quality control, including assessment of library size, mapping rate, and gene count distributions. Low-count genes were filtered before normalisation. Gene expression was normalised using the trimmed mean of M-values (TMM) method in edgeR, followed by log₂ transformation to obtain normalised expression values suitable for downstream analyses.¹⁸

Following platform-specific QC and normalisation, gene expression values were Z-score-normalised within each platform to place microarray intensities and RNA-seq-derived expression on a comparable scale. This approach preserves within-platform relative expression while enabling integrated gene expression analyses across platforms. Z-score transformation has been previously demonstrated as an effective and practical strategy for cross-platform harmonisation of transcriptomic data.^{19–21}

Air pollution estimation

To estimate ambient air pollution exposure for SARP participants, we calculated annual average concentrations of five pollutants: PM_{2.5}, NO₂, PM₁₀, O₃, and SO₂.

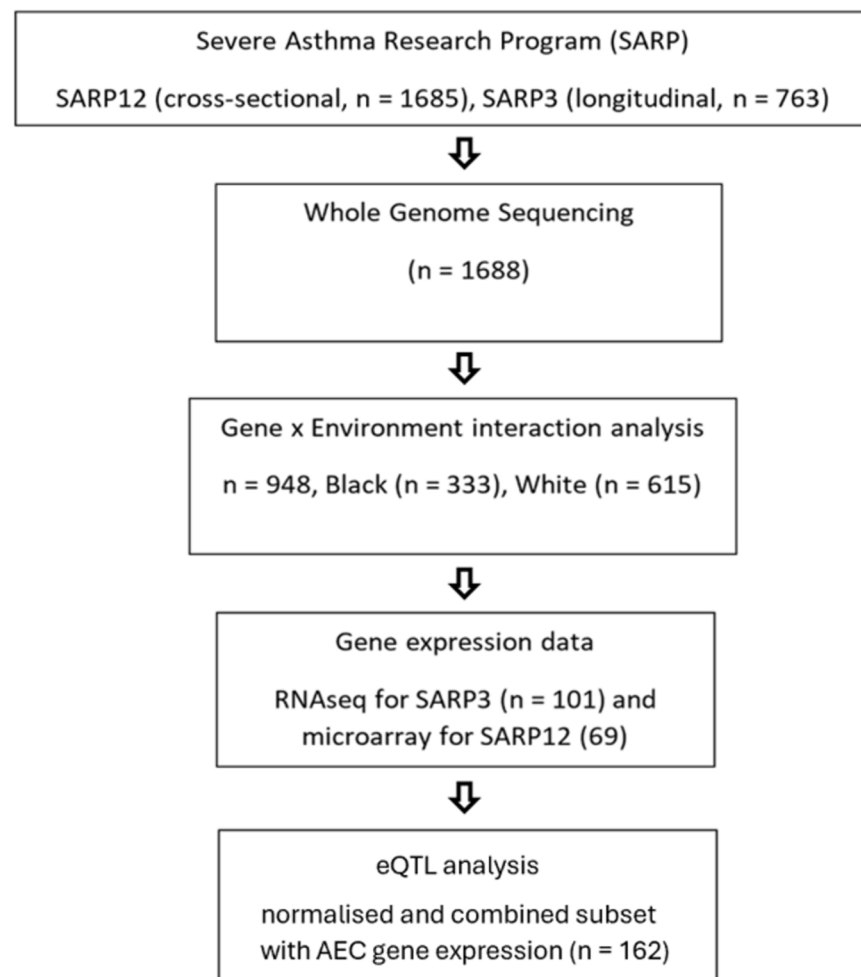


Fig. 1: Study flowchart showing participant selection and exclusions. Participants were drawn from the Severe Asthma Research Program (SARP), including SARP1,2 (cross-sectional, n = 1685) and SARP3 (longitudinal, n = 763). A total of 1688 participants underwent whole genome sequencing. Of these, 948 participants with complete data were included in the GxE interaction analysis, stratified by self-reported race (Black, n = 333; White, n = 615). Gene expression data were available for a subset of participants, comprising RNA sequencing from SARP3 (n = 101) and microarray data from SARP12 (n = 69). For eQTL analyses, normalised and combined airway epithelial cell gene expression data were used, yielding a final subset of n = 162 participants.

The year 2011 was selected as it represents the midpoint of the SARP1–3 recruitment period. Pollutant data were obtained from the U.S. Environmental Protection Agency's Air Quality System (AQS). Exposure levels were assigned to participants based on a nearest-neighbour interpolation method, using the closest monitoring station to the geographic center of each participant's ZIP code. The distance between the ZIP code centroid and the nearest monitor was calculated and categorised into three groups (<3 miles, 3–10 miles, >10 miles), and these distance-based weights were incorporated into subsequent GxE analyses. This approach assigns PM_{2.5} exposure using annual averages at each participant's 5-digit ZIP-code centroid and applies distance-based weighting of monitoring stations

to help reduce spatial misclassification. When a primary monitoring station had incomplete annual data, we assigned exposure using the nearest available monitor within the same distance tier, averaging values across monitors when multiple candidates were available.

Selection of SNPs and eQTL analysis

To examine gene–environment interactions, SNPs in oxidative stress–related genes (436 genes) were prioritised using the GO:0006979 pathway. Only the 120 with available eQTL data in GTEx were selected, as accessed in 2021. SNPs were extracted from WGS database using PLINK 1.9. Expression quantitative trait locus (eQTL) analysis was performed using a linear additive model in

R to associate SNP genotype with gene expression. eQTL analyses were restricted to cis-eQTLs, defined as SNPs located within ± 1 Mb of the transcription start site of the gene. For SNPs showing significant G $\times$ E interactions, we queried the GTEx v8 Human Lung tissue dataset to assess whether the same SNPs demonstrated significant SNP–gene expression associations in lung tissue. Associations with FDR-adjusted $p < 0.05$ were considered evidence of tissue-specific regulatory relevance.

Statistical analysis

The association between pollution exposure and FEV₁% predicted was evaluated using linear regression to estimate the effect size (β) for each pollutant. The standard error of the β coefficient was calculated based on the residual sum of squares from the model and the variance of the exposure variable, with 95% confidence intervals computed using the corresponding t-distribution. Associations between individual pollutants and FEV₁% predicted were visualised in a forest plot, presenting β estimates and their 95% confidence intervals. Gene-environment interactions were assessed using weighted linear regression, which accounted for the distance between the nearest air quality monitor and the participant's ZIP code centroid, to improve exposure estimate precision. Interaction terms between SNPs and pollutant levels were included to quantify the joint effect of genetic variation and environmental exposure on lung function. We first identified SNPs located within ± 50 kb of genes in the oxidative stress response pathway, followed by filtering using a minor allele frequency $> 1\%$ and Hardy–Weinberg equilibrium $p > 1 \times 10^{-6}$. Linkage disequilibrium pruning was conducted at $r^2 < 0.8$ using PLINK. After quality control, 4337 candidate SNPs were retained for gene–environment interaction testing. SNPs were coded additively (0, 1, 2 copies of the effect allele), and PM_{2.5} was treated as a continuous variable. Age, sex, and height are incorporated in the calculation of FEV₁% predicted, and all participants were non-smokers (< 10 pack-years and no smoking in the prior year). Limited socioeconomic variables (e.g., education) were tested but did not materially influence results, and BMI was not included to avoid over-adjustment. Analyses were conducted in the combined cohort. Genetic ancestry was characterised using the TOPMed ancestry pipeline, which computes principal components with PC-AiR and PC-Relate on high-quality, LD-pruned biallelic SNPs, accounting for relatedness and population.^{15,22} Models were adjusted for the top five principal components derived from genetic ancestry analyses to account for potential confounding due to population structure. A directed acyclic graph illustrating the assumed causal relationships among PM_{2.5} exposure, genetic variation, lung function, and key covariates is shown in [Supplementary Figure S1](#). P-values were

derived from two-sided Wald tests of regression coefficients in linear models. For gene–environment analyses, p-values correspond to the SNP $\times$ PM_{2.5} interaction term. For eQTL analyses, p-values were obtained from linear additive models testing genotype–expression associations. To account for multiple comparisons, p-values were adjusted using the Benjamini–Hochberg false discovery rate (FDR) procedure. Associations with FDR-adjusted $p < 0.05$ were considered statistically significant unless otherwise specified. Significant interaction results were visualised in a volcano plot, displaying the β estimates and log₁₀-transformed p-values of the interaction terms for all tested SNPs. Normality of residuals was evaluated using Q–Q plots ([Supplementary Figure S2](#)). Residuals demonstrated approximate normality with minor deviation in the tails, and no major violations of model assumptions were observed. All statistical analyses were conducted using R software (version 4.2.2).

Role of funders

The funding sources had no role in study design, data collection, data analysis, interpretation of data, writing of the manuscript, or the decision to submit for publication. No one was paid to write the article.

Results

Baseline demographics

The starting study population included 2448 patients with asthma from a total of nine different research sites in the US who were recruited into the SARP 1–3 cohorts ([Fig. 1](#)). For inclusion in the analysis, participants were required to have both 5-digit U.S. zip codes and whole genome sequencing data from the TOPMed program. GxE analysis included all participants with available WGS and ZIP code data ($n = 948$). Self-reported race is shown in [Table 1](#) but was not used for stratified analyses, as genetic ancestry was controlled using the top five principal components. Baseline characteristics are presented in [Table 1](#), including sex, race, age, and baseline FEV₁% predicted. Over 200 SARP participants underwent bronchoscopy to obtain epithelial gene expression data (by microarray in SARP1-2 and bulk RNA sequencing in SARP3). However, not all participants had complete zip code or WGS data available ([Fig. 1](#)). The demographics of this subpopulation did not differ significantly from the overall study population, and additional characteristics were detailed in [Table 1](#).

Air pollution exposure levels were higher in industrialised cities

Air pollution exposure was assigned using annual average 2011 concentrations from the nearest EPA monitoring station to each participant's 5-digit ZIP-code centroid. Median exposure levels are

Characteristic	All patients, n = 948	SARP 1-2, n = 386	SARP3, n = 562	RNAseq (BEC), n = 101	Microarray (BEC), n = 69
Age, Median (IQR)	38 (22-52)	33 (21-47)	42 (22-54)	39 (31-52)	33 (23-46)
Sex					
Male, n (%)	366 (39)	150 (39)	216 (38)	35 (35)	20 (29)
Female, n (%)	582 (61)	236 (61)	346 (62)	66 (65)	49 (71)
Race					
Black, n (%)	333 (35)	144 (37)	189 (34)	30 (30)	24 (35)
White, n (%)	615 (65)	242 (63)	373 (66)	71 (70)	45 (65)
Baseline FEV ₁ % predicted, Median (IQR)	79.44 (64.27-93.08)	80.50 (65.25-95.00)	77.40 (62.49-92.07)	77.11 (62.90-91.06)	75.00 (60.00-92.00)

Table 1: Demographics of study participants.

summarised in [Table 2](#). All median exposure levels exceeded EPA and WHO recommended limits. [Table 2](#) also highlights differences in median pollutant exposures between patients in industrialised cities and other locations, showing significantly higher levels of individual pollutants in industrialised areas. The map of PM_{2.5} exposure level at patients' location is in [Supplementary Figure S3](#).

PM_{2.5} associates with lower baseline FEV₁% predicted

To determine the impact of various air pollutants on lung function in patients with asthma, linear regression was conducted on race-neutral baseline FEV₁% predicted (95% confidence interval). Exposure to higher PM_{2.5} associated with a lower FEV₁% predicted [$\beta = -0.7$ (CI: -1.23 to -0.16 , p -value = 0.01)]. Thus, for each 1 $\mu\text{g}/\text{mm}^3$ increase in PM_{2.5}, there is a 0.7% decrease in FEV₁% predicted, with a CI of 0.16% and 1.23% ([Fig. 2](#)). The associations with other pollutants were not significant: SO₂: $\beta = 0.12$, CI -0.14 to 0.37 ; PM₁₀: $\beta = -0.02$, CI -0.21 to 0.17 ; O₃: $\beta = -0.004$, CI -0.14 to 0.13 ; NO₂: $\beta = -0.007$, CI -0.12 to 0.11 .

Gene by environment interactions modify FEV₁% predicted

PM_{2.5}, O₃, SO₂, and NO_x increase oxidative stress, impacting immune responses and mitochondrial function.²³⁻²⁵ Based on this, we selected 120 GO:0006979 oxidative stress pathway genes (with eQTL data), yielding 4337 SNPs after quality control. Weighted linear regression was used to test whether these SNPs modified FEV₁% predicted in response to PM_{2.5} exposure through GxE interactions. [Fig. 3](#) displays volcano plots showing β values for the interaction term (x-axis) and the $-\log_{10}$ transformed p -values after multiple testing corrections (y-axis). Notably, 20 SNPs in 7 genes (*LRRK2*, *MSRA*, *MSRB2*, *OXSRI*, *PXDN*, *TPO* and *APP*) interacted with PM_{2.5} to potentially influence FEV₁% predicted. None of the 20 SNPs were independently associated with FEV₁% predicted (except TPO SNP).

SNPs influenced by PM_{2.5} exposure alter cis-gene expression

To determine whether identified SNPs influence FEV₁% predicted by modulating expression of their corresponding genes in response to PM_{2.5} exposure, we conducted an eQTL analysis. PM_{2.5} was dichotomised at the median (10.3 $\mu\text{g}/\text{m}^3$) due to limited sample size. Analysis included the 20 SNPs showing GxE interactions with FEV₁%. Five SNPs significantly affected both FEV₁% predicted and cis-gene expression in the presence of PM_{2.5} (before correction for multiple comparisons, [Supplementary Table S1](#)). Allele differences at most of these SNPs did not independently affect FEV₁% predicted ([Figs. 4-5](#)). However, the minor "A" allele at the rs2011 locus in *OXSRI* associated with worsening FEV₁% predicted as PM_{2.5} increased ([Fig. 4b](#)). In the smaller subgroup with gene expression data, the A allele at rs2011 in *OXSRI* tended to associate with a lower FEV₁% predicted, particularly in the AA homozygotes when PM_{2.5} exposure was high ([Fig. 4c](#)), consistent with findings in the overall group. These physiologic changes associated with lower expression of *OXSRI* mRNA in AA homozygotes, but only under conditions of high PM_{2.5} exposure ([Fig. 4d](#)). Similarly, rs1330782395 heterozygotes in *PXDN* showed reduced FEV₁% and *PXDN* expression under high PM_{2.5} ([Fig. 4E-H](#)). This suggests that both *OXSRI* and *PXDN* expression protect against redox-related PM_{2.5}-associated injury and that the minor alleles at each of these SNPs limit these biologic responses and worsen FEV₁% predicted.

Although TT (minor allele) homozygotes at rs62107964 of the *TPO* locus exhibited lower FEV₁% predicted at baseline (p -value = 0.02), consistent with their higher *TPO* expression ([Fig. 5a](#) and [d](#)), FEV₁% predicted was higher in TT homozygotes as PM_{2.5} exposure increased, likely due to suppression of *TPO* expression under high pollutant exposure ([Fig. 5b](#) and [d](#)). Similarly, in the subset with gene expression data, FEV₁% predicted was higher in TT homozygotes patients, compared to the other genotypes, but only as PM_{2.5} exposure increased ([Fig. 5c](#)). In the presence of

Atmospheric pollutant	All participants (n = 948)	Industrialised sites (n = 669)	Less industrialised sites (n = 279)	P-value between sites	EPA threshold	WHO threshold
PM _{2.5} (µg/m ³), Median (IQR)	10.02 (9.29–11.16)	11.00 (10.15–12.32)	9.62 (9.15–10.41)	<0.001	9.0 µg/m ³ (annual)	10 µg/m ³ (annual)
NO ₂ (ppb), Median (IQR)	18.94 (11.02–24.83)	18.76 (15.24–22.73)	18.04 (9.8–16.69)	<0.001	28 ppb (annual)	21.25 ppb (annual)
PM ₁₀ (µg/m ³), Median (IQR)	16.57 (14.64–18.37)	17.47 (16.10–23.10)	16.27 (14.47–17.42)	<0.001	150 µg/m ³ (24-h standard)	50 µg/m ³ (24-hour standard)
SO ₂ (ppb), Median (IQR)	1.68 (0.57–2.75)	2.82 (1.95–6.25)	1.13 (0.55–1.85)	<0.001	28 ppb (1-h standard)	7.61 ppb (24-h standard)
O ₃ (ppb), Median (IQR)	46.52 (42.59–50.42)	47.91 (42.64–50.42)	43.55 (41.13–46.38)	<0.001	70 (8-h average)	60 (8-h average)

Median (IQR) pollutant levels in 2011 at participants' ZIP code locations, estimated from the nearest EPA monitor. Values are stratified by industrialised vs less industrialised sites. P-values, calculated from Wilcoxon rank-sum test, indicate between-group differences; EPA and WHO thresholds shown for reference.

Table 2: Ambient air pollutant levels at participants' ZIP code locations.

low PM_{2.5} exposure, TT homozygotes had lower FEV₁% predicted (Fig. 5b–c) and higher *TPO* expression (Fig. 5d) suggesting that *TPO* expression overall negatively influenced lung function. However, when the same genotype was exposed to high PM_{2.5} the expected increase in *TPO* expression was blunted and higher FEV₁% predicted was observed as compared to the other genotypes (Fig. 5b–d), supporting a protective effect of this genotype. Indeed, the T allele at rs62107964 eQTL, in the presence of increasing PM_{2.5} exposure associated with marked inhibition of *TPO* expression, protecting lung function. Similar patterns were seen for *TPO* rs12471599 and *LRRK2* rs73102291 (Supplementary Figures S4–5; Table 3).

Discussion

The findings reported here represent one of the broadest scale gene–environment interactions studies in asthma to date. While this study replicated earlier findings that exposure to PM_{2.5} linked to lower FEV₁%

predicted,²⁶ it significantly expands on those findings by identifying interactions of zip code defined PM_{2.5} exposures with personal genetic sequencing variation, specifically among 120 genes well recognised to modulate intracellular redox balance (GO:0006979), which may affect changes in lung function. Our findings complement a large genome-wide gene–air pollution interaction study in UK Biobank, which identified loci influencing pollution-related lung function in the general population. Unlike that study, we focused on oxidative stress pathways in a well-characterised asthma cohort and integrated airway epithelial gene expression data to provide mechanistic insight.²⁷ 20 SNPs from 7 redox-related genes (*LRRK2*, *MSRA*, *MSRB2*, *OXSRL1*, *PXDN*, *TPO*, and *APP*) were identified for their GxE interactions. Additional gene expression data in a smaller subset identified 5 of these SNPs (in 4 of the 7 genes) as eQTLs affecting cis-gene expression, and particularly under higher PM_{2.5} exposure, the observed effects occurred in biologically meaningful and predictable directions; that is, genetic variants associated

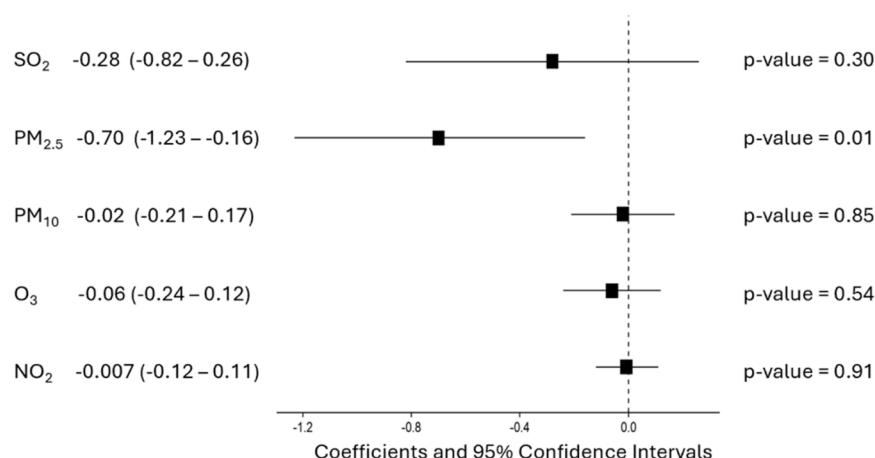


Fig. 2: Forest plot depicting the association between air pollution exposure and FEV₁% predicted in SARP. Effect estimates were presented as beta coefficients with 95% confidence intervals (CI) for each pollution from linear regression models (n = 948). The horizontal lines represented the 95% CI, and the center markers indicated the beta estimates. A vertical dashed line represented the null value (Beta = 0), indicating no association. P-values were calculated using two-sided Wald tests from linear regression models evaluating the association between each pollutant and FEV₁% predicted.



Fig. 3: Volcano plot showing gene-environment interactions on FEV₁% predicted. Each point represents a gene-environment interaction, with the x-axis indicating the effect size and the y-axis representing the log transformed p-value (two-sided Wald tests in weighted linear regression models) for the interaction term after correcting multiple testing. Significant associations are highlighted in blue ($n = 948$, p -value < 0.05) and non-significant associations in red. The horizontal dashed line indicates the threshold for statistical significance ($p < 0.05$). Models were adjusted for principal components for genetic background. Full names of significant genes: APP (Amyloid Precursor Protein), LRRK2 (Leucine-Rich Repeat Kinase 2), MSRA (Methionine Sulfoxide Reductase A), MSRB2 (Methionine Sulfoxide Reductase B2), OXSR1 (Oxidative Stress Responsive Kinase 1), PXDN (Peroxisidase), TPO (Thyroid Peroxidase).

with reduced expression of oxidative stress-related genes under high PM_{2.5} exposure corresponded to expected decreases (or protection) in lung function, consistent with known mechanisms of particulate matter-induced oxidative injury.

Perhaps not surprisingly, we observed higher PM_{2.5} (and various other atmospheric pollutant) levels in the zip codes associated with the industrialised SARP sites as compared to the other centres. Additionally, emitting sources (factories/power plants) in these industrialised cities are likely to be sources for sulfates and heavy metals, which can interact with PM_{2.5} to increase its toxicity and the risk for respiratory diseases.²⁸ However, like cigarette smoking, environmental exposure to atmospheric pollutants can have varying effects dependent on each individual's genetics. Indeed, atmospheric pollution is widely believed to have a greater clinical

impact on patients with respiratory diseases.⁴ Therefore, a deeper understanding of the complex interactions of the environment with the genome is needed.

Twin and family-based studies estimate that heritability of lung function ranges from 26% to 81%, highlighting the contribution of genetic factors to FEV₁.²⁹ However, genetic studies have traditionally overlooked the role of environmental factors in asthma. Most existing gene-environment interaction studies in asthma are limited to a small set of candidate genes, particularly the glutathione-S transferase (GST) family,^{30,31} and have primarily focused on susceptibility to childhood onset asthma. For example, studies have shown that the risk of childhood-onset asthma varies based on the interaction of *GSTM1* genotype with maternal smoking during pregnancy and

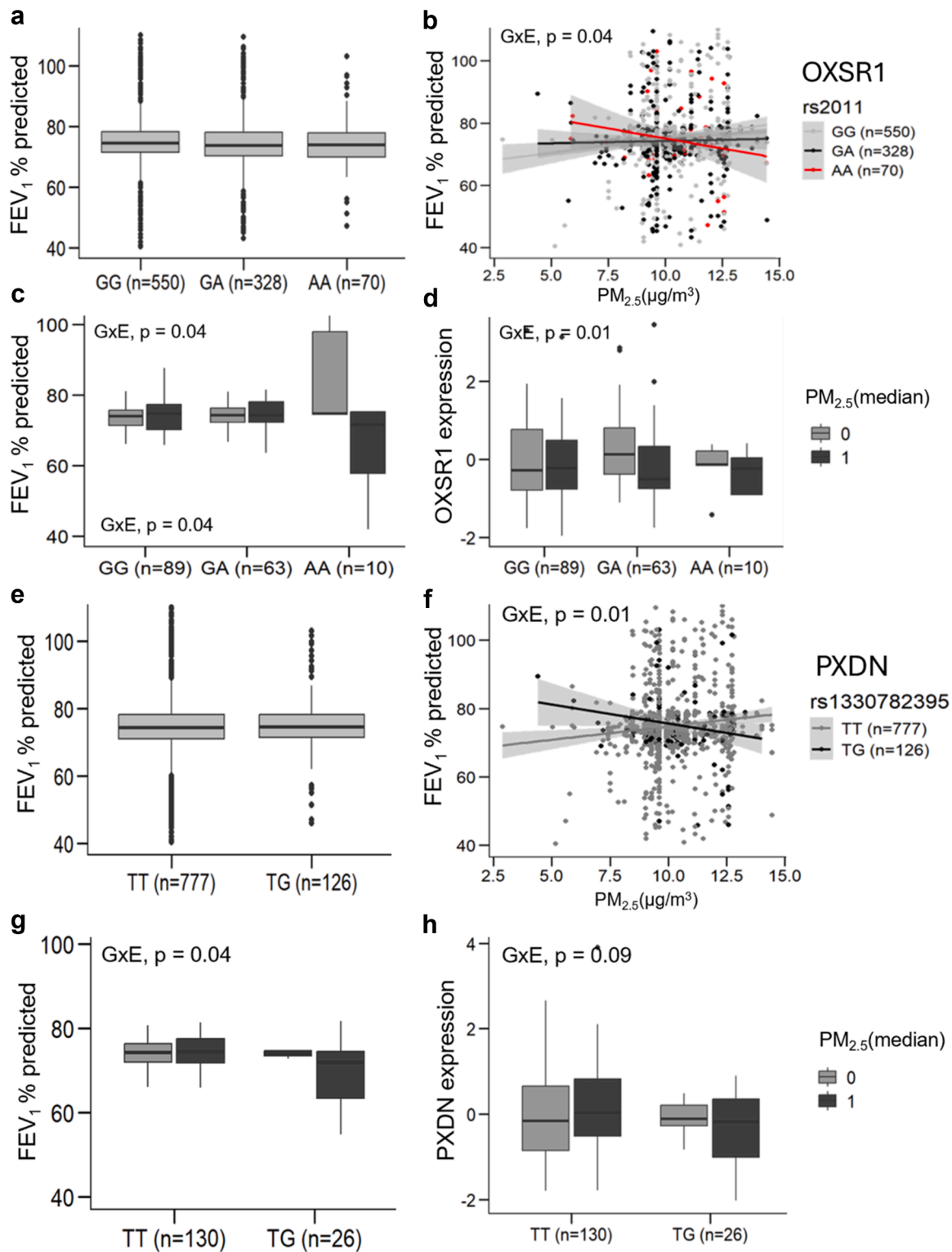


Fig. 4: Interaction of OXSR1 and PXDN genotypes with PM_{2.5} exposure on lung function and gene expression. a–d: Association of OXSR1 rs2011 genotype and PM_{2.5} exposure with lung function (n = 948) and gene expression (n = 162). (a) Baseline FEV₁% predicted by rs2011 genotype alone is not different across genotypes. (b) Interaction of rs2011 genotype with levels of PM_{2.5} exposure. AA homozygotes have a sharper decline in FEV₁% predicted with increasing PM_{2.5} exposure compared to other genotypes. (c) In the gene expression subset, AA

environmental tobacco smoke exposure (ETS) in childhood.³² However, few studies have performed comprehensive GxE analyses focused on lung function.³³

Major challenges in conducting GxE studies include the lack of large geocoded populations with WGS. Of 30 GxE studies in asthma, only 9 included more than 500 subjects.³⁴ Although the SARP population started with over 2400 patients, after limitations from WGS and zip code data availability, <1000 patients were included in the final GxE analysis. Yet this still represents one of the largest GxE studies to date in asthma, allowing us to explore an expanded candidate gene approach. Even this larger sample size is grossly underpowered to detect GxE interactions across the genome that survive multiple comparison testing. Although oxidative stress is a well-established mechanism of PM_{2.5}-related injury, air pollutants also affect epithelial barrier function, immune signalling, and epigenetic regulation. As the small sample size of this study drastically limited power for a genome-wide agnostic GxE scan, we used a supervised pathway approach, examining thousands of variants across 120 oxidative stress-related genes. This expands upon traditional targeted studies of single genes while maintaining statistical power and enabling biologically interpretable findings, while leaving open opportunities to examine other pathways. Future studies in larger cohorts (or targeted analysis of other pathways) will be needed to evaluate genome-wide GxE signals and to investigate additional pathways beyond oxidative stress. Thus, we chose a candidate pathway approach, focussing on 120 oxidative stress-related genes to compare SNPs, their relation to FEV₁% predicted and the modification by PM_{2.5} exposure. Including the top five principal components as covariates accounts for subtle differences in genetic ancestry across participants, allowing analyses in the combined cohort without race stratification.

This study identified 20 SNPs in 7 oxidative stress-related genes that appear to interact with PM_{2.5} to influence lung function in patients with asthma. These genes can be broadly and biologically categorised as potentially beneficial (antioxidant) or harmful (oxidative). One of the genes (with 4 identified SNPs) is thyroid peroxidase (*TPO*). *TPO* encodes a haem-containing

peroxidase that catalyses the oxidation of substrates by hydrogen peroxide, playing a critical role in thyroid hormone synthesis.³⁵ Importantly, *TPO* expression has been reported in the lung, where it could contribute to the formation of ROS and oxygen radicals which worsen oxidative stress.³⁵ Thus, although everyone carries this genetic locus, the impact on lung function is modified by the specific allele present at each site; notably, the minor T alleles of *TPO* SNPs (rs62107964 and rs12471599) appear to interact with PM_{2.5} levels with a dampening of *TPO* expression, preserving lung function despite increasing PM_{2.5} exposure.

A similar trend was observed with leucine-rich repeat kinase 2 (*LRRK2*) SNP (rs73102291). *LRRK2* contributes to ROS production, causing mitochondrial dysfunction, apoptosis, and impaired cellular integrity under oxidative stress.^{36,37} Although the A allele was not associated with baseline lung function, it correlated with lower *LRRK2* expression and improved lung function under high PM_{2.5} exposure. These findings suggest *LRRK2* and *TPO* may be potential therapeutic targets in asthma, particularly in individuals with high oxidative stress and pollutant exposure.

In contrast, *OXSRI* (Oxidative Stress Responsive Kinase 1) is recognised for its role in antioxidant stress-response pathways, whereas *PXDN* (Peroxidasin) functions as a haem peroxidase in the extracellular matrix.^{38,39} *OXSRI* facilitates cellular adaptation to oxidative stress by regulating antioxidant defences,³⁹ while *PXDN* catalyses collagen IV cross-linking through a haem peroxidase-mediated reaction within the basement membrane.^{40,41} Increased expression of these proteins likely enhances the cell's ability to adapt to oxidative stress (*OXSRI*), maintain extracellular matrix integrity (*PXDN*) and maintain lung function, especially under environmental challenges like PM_{2.5} exposure. Importantly, genetic variation affects this protection: the *OXSRI* rs1384006 T allele is linked to asthma exacerbations, reduced expression, and weaker defences.³⁹ Although specific *PXDN* SNPs are less defined, certain variants similarly reduce expression and lung function, potentially impairing basement membrane stability.^{40,42} Together, these findings emphasise *OXSRI* and *PXDN* as key protective factors against pollutant-induced oxidative stress in asthma,

homozygotes exhibit lower FEV₁% predicted under high PM_{2.5} exposure compared to other genotypes. (d) mRNA expression is lower in AA homozygotes at the rs2011 genotype under conditions of high PM_{2.5} exposure, compared to other genotypes. e–h: Association of *PXDN* rs1330782395 genotype and PM_{2.5} exposure with lung function (n = 948) and gene expression (n = 162). (e) Baseline FEV₁% predicted by rs1330782395 genotype alone is not different across genotypes. (f) Interaction of rs1330782395 genotype with levels of PM_{2.5}: heterozygotes (TG) exhibit a steeper decline in FEV₁% predicted with increasing PM_{2.5} exposure compared to other genotypes. (g) In the gene expression subset, TG heterozygotes also exhibit lower FEV₁% predicted under high PM_{2.5} exposure compared to other genotypes. (h) *PXDN* mRNA expression by rs1330782395 genotype under high PM_{2.5} exposure: *PXDN* mRNA expression is lower in the TG heterozygotes at rs1330782395 under conditions of high PM_{2.5} exposure, compared to other genotypes. Error bars represent standard error. P-values for genotype and genotype × PM_{2.5} interaction effects were obtained from linear regression models including the corresponding main and interaction terms. For gene expression analyses, p-values were derived from linear additive eQTL models.

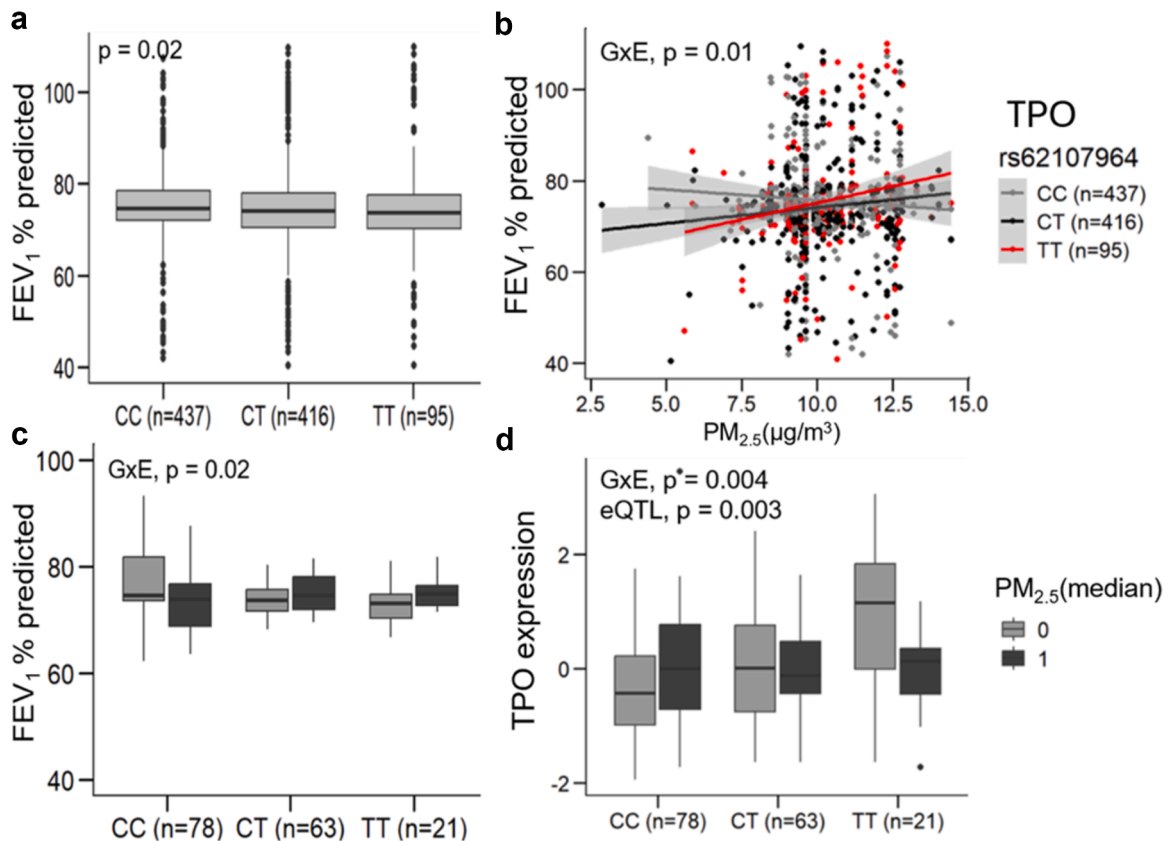


Fig. 5: PM_{2.5} exposure reverses the effect of TPO rs62107964 genotype on lung function and gene expression. (a) Baseline FEV₁% predicted by rs62107964 genotype: TT homozygotes have lower FEV₁% predicted irrespective of PM_{2.5} exposure (n = 948), $p = 0.02$. (b) Interaction of rs62107964 genotype with PM_{2.5}: TT homozygotes exhibit higher FEV₁% predicted with increasing PM_{2.5} exposure as compared to other genotypes (n = 948). (c) In the gene expression subset, TT homozygotes exhibit higher FEV₁% predicted under high PM_{2.5} exposure compared to other genotypes. (d) TPO mRNA expression by rs62107964 genotype under PM_{2.5} exposure: At low PM_{2.5} exposure, TT homozygotes have higher TPO expression (eQTL, n = 162, $p = 0.003$) which is protectively suppressed under high PM_{2.5} exposure compared to other genotypes ($G \times E, p = 0.004$). Some genotype strata showed qualitative differences in the direction of association; therefore these results should be interpreted cautiously and considered hypothesis-generating. Error bars represent standard error. P-values for genotype and genotype $\times$ PM_{2.5} interaction effects were obtained from linear regression models including the corresponding main and interaction terms. For gene expression analyses, p-values were derived from linear additive eQTL models.

suggesting that boosting their stress-response and matrix-stabilising functions could help mitigate oxidative damage and improve clinical outcomes.

Similar gene–environment interactions with PM_{2.5} exposure and lung function were identified for SNPs in *APP* (amyloid precursor protein), *MSRA* (methionine

Gene	SNP	Chr	Position (hg38)	Risk genotypes alone effect gene expression	eQTL under exposure (beta and p-value) ^a	
					beta	p-value
<i>OXSRI</i>	rs2011	3	38,255,326	No	−0.68	0.01
<i>LRRK2</i>	rs73102291	12	40,213,708	No	−0.96	0.02
<i>PXDN</i>	rs1330782395	2	1,720,069	Yes (trend to decrease)	−0.20	0.09
<i>TPO</i>	rs12471599	2	1,479,112	Yes (increase)	−0.81	0.04
	rs62107964	2	1,480,427	Yes (increase)	−0.65	0.003

^acis-eQTL associations were evaluated using linear regression models with gene expression as the outcome and SNP genotype as an additive predictor. P-values were obtained from two-sided Wald tests of the regression coefficients under exposure-stratified or exposure-conditioned models.

Table 3: GxE on cis-gene expression in identified SNPs.

sulfoxide reductase A), and *MSRB2* (methionine sulf-oxide reductase B2), all of which play key roles in oxidative stress response.^{43–45} However, their expression in airway epithelial cells was not affected, suggesting that expression changes in other asthma-relevant cell types, such as immune or structural cells, may be involved. While the interaction p-values are modest, the consistent directionality across multiple SNPs and concordant eQTL findings support the biological relevance of these novel PM_{2.5}-gene interactions. Notably, several SNPs showed qualitative interaction patterns, where minor-allele carriers exhibited opposite directionality relative to the reference genotype. Given the modest sample size and risk of false-positive qualitative interactions, these findings should be interpreted cautiously and viewed as hypothesis-generating. Future studies with larger samples and independent cohorts will be needed to validate these observations. Although the SNPs we identified all reside within oxidative stress-related pathways, the directions of interaction effects were not uniform. This variation is biologically plausible because oxidative stress pathways include both antioxidant and pro-oxidant processes, and individual genes have distinct, sometimes opposing mechanistic roles. As shown in Figs. 4 and 5, reduced expression of antioxidant genes such as *OXSRI* and *PXDN* under higher PM_{2.5} exposure was associated with lower FEV₁, whereas reduced expression of the pro-oxidant enzyme *TPO* was associated with improved lung function. These context-dependent, pathway-consistent effects support biologically coherent mechanisms rather than random pleiotropy or analytic artifact.

Other pollutants, including PM₁₀, did not show significant associations with FEV₁% predicted. PM₁₀ was only moderately correlated with PM_{2.5} in our dataset ($r \approx 0.47$, [Supplementary Figure S6](#)) and represents total particulate mass $\leq 10 \mu\text{m}$, which includes but does not isolate the fine fraction. PM_{2.5}, with its small ($2.5 \mu\text{m}$ size or less), is more likely to impact bronchial airways and beyond. It has also been more strongly linked to redox changes, small-airway pathology and disease.⁴⁶ This pattern is consistent with prior literature where PM_{2.5} demonstrates stronger associations with asthma morbidity than PM₁₀.

The results of this study are limited by the imprecision of individual-level environmental exposures. Assigning exposure at the ZIP-code level introduces spatial imprecision, particularly in areas with heterogeneous PM_{2.5} distributions, but this misclassification is expected to be non-differential and likely biases results toward the null. Participants were localised to 5-digit U.S. ZIP codes, which cover broad geographic areas—averaging 82.25 square miles. Distance-based weighting from the nearest air quality monitor estimated PM_{2.5} levels, introducing potential spatial error of up to 5 miles.⁴⁷ Notably, previous research has

employed buffer radii ranging from 6.4 km to 50 km around monitors, supporting the validity of our approach within the commonly accepted spatial range in air pollution epidemiology.⁴⁸ We also note that this ZIP-code/monitor-based assignment induces primarily Berkson-type spatial measurement error, in which individuals' true exposures vary around the assigned ZIP-level mean. This form of non-differential error with respect to genotype is expected to attenuate exposure-response and GxE estimates toward the null and reduce statistical power rather than create spurious interactions. To address temporal misclassification, we repeated the analyses using average annual PM_{2.5} for both 2008 and 2015. β estimates remained directionally consistent, with smaller p-values in 2008 (higher PM_{2.5} levels) and larger p-values in 2015 (lower PM_{2.5} levels), supporting a longer-term, more chronic effect of PM_{2.5} rather than acute results driven by a single exposure year ([Supplementary Table S2](#)). These results may reflect the nature of lung function in adult asthma where variations from day to day or even year to year are modest.⁴⁹ Despite these limitations, the strengths of our study include a novel approach to identifying biologically meaningful gene-environment interactions, which encompass over 400 oxidative stress pathway genes, that may worsen asthma in genetically susceptible individuals.

In addition to exposure limitations, our findings may be confounded by race and socioeconomic status (SES). While we stratified analyses by racial group (Black and White), residual differences in genetic ancestry, exposure, and structural factors may still bias results. PM_{2.5} exposure is disproportionately higher in racially and economically marginalised communities.⁵⁰ Excluding SNPs with discordant effects between groups may improve reliability but limits generalisability. The absence of individual-level SES data, such as income and education, and reliance on area-level proxies may obscure within-group differences.⁵⁰ Healthcare disparities could also confound associations between pollution, genetics, and lung function. Future studies with refined ancestry, exposure, SES, and treatment data are needed to better understand these complex, intersecting influences.

In conclusion, our study represents one of the largest GxE studies to date in asthma and the first to integrate airway gene expression data. Our approach expands previous GxE research in asthma by analysing thousands of known SNPs in 120 oxidative stress-related genes. Using this approach, we identified 20 SNPs across seven oxidative stress-related genes that interact with PM_{2.5} exposure to influence asthmatic lung function and airway gene expression. These findings could begin provide critical insights into the mechanisms behind the effects of pollution on human health while identifying novel therapeutic targets for individuals at high risk due to PM_{2.5} exposure.

Contributors

Shuangjia Xue developed the hypothesis and study design, analysed the data and developed the code, wrote the manuscript.

Sayed Mehdi Nouraie assisted in procuring and analysing the pollution data, advised on statistical analysis, and edited the manuscript.

Jiebiao Wang advised on data analysis.

Xingnan Li assisted with access to TOPMed data and sharing.

Deborah Meyers assisted with access to TOPMed data and advised on analytics, and reviewed the manuscript.

Eugene R Bleecker assisted with access to TOPMed data and advised on analytics reviewed the manuscript.

Victor E. Ortega advised on analytics reviewed the manuscript.

Mario Castro recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Loren C. Denlinger recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Serpil Erzurum Recruited the participants and reviewed the manuscript.

Elliot Israel recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Prescott Woodruff contributed to the RNAsequencing data, recruited participants, and performed bronchoscopies.

Wendy C Moore recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Stephanie Christenson performed the RNAsequencing.

John Fahy recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Kaharu Sumino recruited the participants.

Nizar Jarjour recruited the participants, performed the epithelial brushings and reviewed the manuscript.

Bruce Levy (Director of the SARP steering committee) reviewed the manuscript.

Sally E Wenzel helped develop the hypothesis and study design, performed the epithelial brushings, helped to perform analyses, wrote and edited the manuscript.

All authors had full access to all the data in the study, reviewed the final version of the manuscript, and approved it for submission. Dr. Xue and Dr. Nouraie accessed and verified all the underlying data.

Data sharing statement

Ground-level pollutant measurements used to estimate exposure were obtained from the U.S. Environmental Protection Agency's Air Quality System (AQS) database. Whole genome sequencing data generated and analysed in this study are available from the NHLBI Trans-Omics for Precision Medicine (TOPMed) program through the database of Genotypes and Phenotypes (dbGaP) under accession number phs001446.v3.p2. Access is granted to qualified researchers under NIH data use agreements. Bronchial epithelial gene expression datasets are publicly available in the Gene Expression Omnibus under accession numbers GSE63142 and GSE43696. Custom R scripts for data processing and statistical analyses are openly available at <https://github.com/shuangjia93/Gene-by-environment-interaction-in-asthma>. All other data supporting the findings of this study are available from the corresponding author on reasonable request, subject to institutional and ethical approvals.

Declaration of interests

Shuangjia Xue, Sayed Mehdi Nouraie, Jiebiao Wang, and Xingnan Li, Deborah Meyers, and Serpil Erzurum declare no conflicts of interest. The following authors declared conflicts the following conflicts unrelated to the current manuscript. Eugene R. Bleecker received grants/contracts from AstraZeneca, Sanofi Genzyme, and TEVA (paid to institution) and NIH NHLBI (SPIROMICS, SARP, EPIPHANY R01, PrecISE; paid to institution); and consulting fees from AstraZeneca. Victor E. Ortega received grants/contracts from NHLBI; participated on independent data monitoring boards for Regeneron and Sanofi and an advisory board for AstraZeneca; and serves as associate editor for JAMA, UpToDate chapter author (asthma genetics), and author for

Merck (asthma). Mario Castro received grants/contracts from the American Lung Association, AstraZeneca, Gala Therapeutics, Genentech, GSK, NIH, Nacion, Novartis, PCORI, Pulmatrix, Sanofi-Aventis, and Regeneron; consulting fees from Allakos, Amgen, Apogee, Apreo Health, Arrowhead Pharmaceuticals, AstraZeneca, Blueprint Medicines, Connect BioPharma, Enveda, Evommune, Generate Biomedicines, GSK, Jasper Therapeutics, Kinaset, Kymera, Novartis, Om Pharma, Pfizer, Pioneering Medicines, Polarean, Regeneron, Sanofi-Aventis, Syneos, Teva, Third Rock Ventures, Uniquity Bio, Upstream Bio, and Verona Pharmaceuticals; payment/honoraria from Amgen, AstraZeneca, MedLearningGroup, Regeneron, and Sanofi-Aventis; and stock in Aer Therapeutics (less than \$5 USD value). Loren C. Denlinger received grants/contracts from NHLBI (PrecISE, GRAIL trials), ALA-ACRC (Lung Health Cohort), and PCORI (iCARE trial); study drug support for the PrecISE trial from GlaxoSmithKline, Laurel, Sun Pharma, Vifor/OM Pharma, Vitaeris/CSL Behring, and Vitaflor; consulting fees from OM Pharma; and honoraria from AstraZeneca; additionally, AstraZeneca, Boehringer-Ingelheim, Genentech, GlaxoSmithKline, Sanofi-Genzyme-Regeneron, and TEVA provided financial support for SARP study activities beyond year three of patient follow-up (these companies had no role in study design or data analysis). Elliot Israel received grants/contracts from AstraZeneca, NIH, PCORI, Sanofi, and TEVA (all paid to institution); royalties from UpToDate (Wolters Kluwer); consulting fees from Amgen, AnaptysBio, Apogee Therapeutics, Arrowhead Pharmaceuticals, AstraZeneca, Bain Capital, Chiesi, Clearview Health Partners, Danker and Associates, Ecor1, ERS, Generate Biomedicine, GlaxoSmithKline, Guidepoint, Jasper Therapeutics, Leerink Partners, Merck, Mountainfield, OrbiMed, Reach Market Research, Regeneron, Sanofi, TEVA, The Lancet, Third Rock Ventures, Windrose Consulting Group, and Yuhan (all paid to him); honoraria from AAN, AFI, AstraZeneca, and Regeneron; payment for expert testimony from Cambridge Medical Experts; travel support from AstraZeneca, Sanofi, and NYU Asthma Airways Symposium; advisory board fees from Amgen, AnaptysBio, Apogee Therapeutics, AstraZeneca, Chiesi, and Sanofi; receipt of equipment, drugs, and materials from Circassia, Genentech, TEVA, Sun Pharma, Laurel Pharmaceuticals, Om Pharma, CSL Behring, Amgen, GlaxoSmithKline, and Sanofi-Regeneron (for various NIH-funded studies); and stock options in Verso (unpaid). Prescott Woodruff received grant support from NIH (SARP); consulting fees from AnaptysBio, AbbVie, and Roche; and honoraria for visiting professorships from BWH, Pitt, and UNC. Wendy C. Moore received grant support from NHLBI (SARP program; paid to institution) and grants/contracts from NHLBI (SARP, PrecISE), ALA (ALA-ACRC Lung Health Cohort), and AstraZeneca, GlaxoSmithKline, Aerteia, Amgen, and TEVA (multicenter clinical trials; paid to institution); and consulting fees from AstraZeneca (advisory board; personal fees). Stephanie Christenson received grants/contracts from NIH, the American Lung Association, COPD Foundation, Department of Defence, and Uniquity Bio (all paid to institution); consulting fees from Sanofi, AstraZeneca, AbbVie, Apogee Therapeutics, and Genentech (all paid to her); honoraria for lectures and presentations from Sanofi, Regeneron, Horizon CME, AstraZeneca, MJH Holdings, WebMD, UpToDate, HCPLive, Answers in CME, PeerView CME, and ReachMD (all paid to her); travel support from Sanofi/Regeneron and GlaxoSmithKline; advisory board fees from Sanofi, AstraZeneca, Regeneron, Illuminate Health, Verona Pharma, Amgen, Uniquity Bio/Devpro Pharma, Kymera Therapeutics, GSK, Genentech, and Gilead (all paid to her); and an unpaid leadership role with the American Thoracic Society. John Fahy received grants/contracts from NIH/NHLBI; consulting fees from Suzhou Connect Biopharmaceuticals, Ltd. and Aer Therapeutics; patents PCT/US2014/028656 and PCT/US19/50475 (thiol-modified saccharides as drugs to treat mucus-associated lung disease); and stock options from Suzhou Connect Biopharmaceuticals, Ltd. and a stock grant from Aer Therapeutics. Nizar Jarjour received consulting fees from GSK, Chiesi, AstraZeneca, and General Medicines. Bruce Levy received grant support from NHLBI, AstraZeneca, Amgen, Genentech, GlaxoSmithKline, Novartis, Sanofi-Genzyme-Regeneron, and TEVA (all paid to institution for SARP) and SRA (paid to institution); royalties from Propeller

Health; and stock options from Noción Therapeutics. Sally E. Wenzel received grant support from NIH/NIAID/NHLBI (paid to institution) and Regeneron (investigator-initiated study); consulting fees from Regeneron, Chiesi, and Eli Lilly (paid to her); and stock in Aer Therapeutics (paid to her and her spouse). Kaharu Sumino received honoraria for a lecture from Kyorin pharmaceutical, support for attending Japan Respiratory Society meeting 2024, participates on the Data Safety and Monitoring Board at The University of Chicago Illinois for ALOHA trial funded by NIH, and is a member of the guideline panel for Joint Task force for American Academy of Allergy, Asthma & Immunology and American College of Allergy, Asthma and Immunology, Guideline for Management of difficult-to-control and severe asthma. Serpil Erzurum has a leadership or fiduciary role in Cleveland Clinic.

Acknowledgements

This study was funded by NIH (HL145406, U01 HL146002, U10 HL109152, U10 HL109168, U10 HL109172, U10 HL109164, U10 HL109086, U10 HL109257, U10 HL109146, and U10 HL109250). The authors acknowledge NHLBI for their longstanding funding of the SARP program, as well as NIAID for their support of Sally Wenzel to evaluate oxidative stress pathways in asthma, the broad focus of this study. The authors would like to thank the SARP clinical coordinators and laboratory personnel for their contributions to fulfilling the scientific mission of the Severe Asthma Research Program. These research findings would not have been possible without their dedication and support in conducting study visits, collecting data, and processing biological samples. The authors would like to thank Dave Mauger and Brenda Phillips, at the SARP data coordinating center for providing the SARP clinical data used in this study. The authors would also like to acknowledge Thomas Brannon for his assistance with the SO₂ data.

Whole genome sequencing for this project was provided by the NHLBI Trans-Omics for Precision Medicine (TOPMed) program (dbGaP accession: phs001446.v3.p2). The authors thank the TOPMed program and the participating studies and participants for generating and sharing genomic data.

Gene expression data were accessed from publicly available repositories (GSE63142, GSE43696). We thank the investigators and participants who contributed to the generation and availability of these datasets.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.ebiom.2026.106334>.

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